SIMATS ENGINEERING

CSA0705 - COMPUTER NETWORKS

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NATURE OF ASSIGNMENT : INDIVIDUAL

1. Given Values and Assumptions

The analysis below uses only the values supplied in the assignment. No value has been changed.

Parameter	Value
Ethernet destination MAC	00:1d:9c:c7:b0:41
Ethernet source MAC	3c:52:82:4e:11:07
EtherType	0x0800
Source IP address	10.10.1.25
Destination IP address	10.10.9.4
TTL (as received)	61
Initial TTL	64
IP protocol field	6 (TCP)
Source port	51522
Destination port	502 (Modbus)
Sequence number	1201
Acknowledgement number	88
TCP data length (robot command)	12 bytes
Camera packet data	1,400 bytes
Ethernet FCS	4 bytes
Industrial EtherType	0x8892
Link speed	100 Mbps

Assumptions used for the calculations (standard header sizes, no options):

Item	Size	Reason
Ethernet II header	14 bytes	6 (dest MAC) + 6 (source MAC) + 2 (EtherType)
IPv4 header	20 bytes	Minimum header, no IP options
TCP header	20 bytes	Minimum header, no TCP options
Ethernet FCS	4 bytes	Given in the assignment

- The 1,400-byte camera packet is treated as **application payload** (data above TCP), the same way the 12-byte robot command is treated.
- 1 Mbps = 1,000,000 bits per second, and 1 byte = 8 bits.
- The values in this report are the **values given in the assignment**. They are not taken from a capture made by the student (see Section 10).

2. Layer-by-Layer Analysis of the Robot Command

When the robot command is sent, each layer adds its own header (and Ethernet also adds a trailer). This is called **encapsulation**. The frame is read from the outside in: Ethernet first, then IPv4, then TCP, and finally the data.

2.1 OSI layers, protocols and PDUs

OSI layer	Protocol / item in this frame	PDU name	Key information from the assignment
7 – Application	Modbus (destination port 502)	Data / message	12-byte robot command
6 – Presentation	No separate header in this frame	Data	Data format is handled inside the application data
5 – Session	No separate header in this frame	Data	Connection is identified by the TCP ports and numbers
4 – Transport	TCP	Segment	Ports 51522 → 502, Seq = 1201, Ack = 88
3 – Network	IPv4	Packet	10.10.1.25 → 10.10.9.4, TTL = 61, Protocol = 6
2 – Data Link	Ethernet II	Frame	MAC addresses, EtherType 0x0800, FCS (4 bytes)
1 – Physical	Cable / signals	Bits	Frame is sent as a stream of bits

2.2 Layer 2 – Ethernet II header (14 bytes)

Field	Value	Size	Meaning
Destination MAC	00:1d:9c:c7:b0:41	6 bytes	Hardware address of the receiving device on this link (unicast)
Source MAC	3c:52:82:4e:11:07	6 bytes	Hardware address of the sending device (unicast)
EtherType	0x0800	2 bytes	Tells the receiver that an IPv4 packet follows
FCS (trailer)	–	4 bytes	Frame Check Sequence; detects bit errors, placed at the end of the frame

2.3 Layer 3 – IPv4 header (20 bytes)

Field	Value	Meaning
Source IP	10.10.1.25	Address of the sender
Destination IP	10.10.9.4	Address of the receiver. It is a different network (10.10.9.x), so routers are needed
TTL	61	Time To Live; each router reduces it by 1
Protocol	6	Upper-layer protocol is TCP
IP packet size	52 bytes	20 (IP header) + 20 (TCP header) + 12 (data)

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2.4 Layer 4 – TCP header (20 bytes)

Field	Value	Meaning
Source port	51522	Port used by the sending application (a temporary client port)
Destination port	502	Well-known Modbus port, so the data is Modbus
Sequence number	1201	Number of the first data byte in this segment
Acknowledgement number	88	Next byte the sender expects to receive from the other device
Data length	12 bytes	Size of the robot command
TCP segment size	32 bytes	20 (TCP header) + 12 (data)

Since 12 bytes of data are carried starting at sequence number 1201, the next byte this sender would use is 1201 + 12 = **1213**.

2.5 Layer 7 – Application data

Destination port 502 identifies the data as a Modbus command. The command is **12 bytes** long. These 12 bytes are the actual useful information; everything else in the frame is overhead added by the lower layers.

2.6 Encapsulation summary

Layer	What is added	Size	Running size
Application	Robot command (data)	12 bytes	12 bytes
Transport	TCP header	20 bytes	32 bytes (segment)
Network	IPv4 header	20 bytes	52 bytes (packet)
Data Link	Ethernet header	14 bytes	66 bytes (frame without FCS)
Data Link	Ethernet FCS	4 bytes	70 bytes (frame with FCS)
Physical	Sent as bits	70 × 8 = 560 bits	–

3. Routers Crossed Using TTL

The sender starts the packet with an initial TTL of 64. Every router that forwards the packet reduces TTL by exactly 1. So the number of routers crossed equals the drop in TTL.

$$\text{Routers crossed} = \text{Initial TTL} - \text{Received TTL}$$
$$= 64 - 61 = 3$$

Answer: The packet crossed 3 routers.

This agrees with the different network numbers of the source (10.10.1.x) and destination (10.10.9.x) addresses.

4. Total Bytes on the Wire (Robot Command Frame)

Total frame = Ethernet header + IPv4 header + TCP header + Data + FCS
= 14 + 20 + 20 + 12 + 4
= 70 bytes

Answer: Total bytes on the wire = 70 bytes (560 bits)

- The frame is longer than the 64-byte minimum Ethernet frame, so no padding is needed.
- Wireshark normally shows **66 bytes** for this frame because the network card usually removes the 4-byte FCS before the capture. 66 + 4 (FCS) = 70 bytes on the wire.

5. Overhead Percentage

Overhead is every byte that is not application data. It is expressed as a percentage of the total frame size.

Overhead = Ethernet + IPv4 + TCP + FCS = 14 + 20 + 20 + 4 = 58 bytes
Overhead % = (Overhead ÷ Total frame) × 100

5.1 Robot command (12 bytes of data)

Total frame = 70 bytes
Overhead % = (58 ÷ 70) × 100 = 82.86 %

Robot command overhead ≈ 82.86 %

5.2 Camera packet (1,400 bytes of data)

Total frame = 14 + 20 + 20 + 1400 + 4 = 1458 bytes
Overhead % = (58 ÷ 1458) × 100 = 3.98 %

Camera packet overhead ≈ 3.98 %

The 1,400-byte camera data is treated as application payload. The IP packet size is 20 + 20 + 1400 = 1440 bytes, which fits inside a standard 1,500-byte Ethernet payload.

5.3 Comparison

Item	Robot command	Camera packet
Application data	12 bytes	1,400 bytes
Overhead (14 + 20 + 20 + 4)	58 bytes	58 bytes
Total frame on the wire	70 bytes	1,458 bytes
Overhead percentage	**82.86 %**	**3.98 %**
Useful data percentage	17.14 %	96.02 %

5.4 What the comparison tells us about small commands

- The headers cost the same **58 bytes** for every frame, whatever the data size.
- For the 12-byte command, more than **four out of five** bytes on the wire are overhead. For the camera packet, less than 4 % is overhead. The command's overhead share is about 20.8 times larger (82.86 ÷ 3.98).
- So small commands use the network inefficiently. When a controller sends many small commands every second, most of the bandwidth is spent on headers, and each frame also needs processing at every layer.
- Large data packets (such as camera images) use the same headers far more efficiently.

6. Mapping Network Faults to OSI Layers

A fault is placed at the layer whose job is not being done correctly. Typical examples are given below.

Network fault	Responsible OSI layer	Reason
Cable unplugged or broken; no link light	Layer 1 – Physical	Signals cannot travel on the medium
Frames dropped because of FCS / CRC errors	Layer 2 – Data Link	FCS checking and framing belong to Ethernet
Wrong destination MAC; switch forwarding problem	Layer 2 – Data Link	MAC addressing is a Layer 2 function
Wrong IP address or wrong gateway; no route to 10.10.9.4	Layer 3 – Network	IP addressing and routing belong to Layer 3
Packet dropped because TTL reached 0	Layer 3 – Network	TTL is an IPv4 field
Port 502 closed; TCP retransmissions; connection refused	Layer 4 – Transport	Ports and reliable delivery belong to TCP
Connection or dialogue between devices ends unexpectedly	Layer 5 – Session	Session control manages the dialogue
Data cannot be understood because of different encoding or encryption	Layer 6 – Presentation	Data format and encryption are Layer 6 functions
Modbus command or function code not accepted by the device	Layer 7 – Application	The application protocol itself is the problem

7. Mapping of the Seven OSI Layers to the Four TCP/IP Layers

OSI layer	TCP/IP layer	Example in this frame
7 – Application	**Application**	Modbus
6 – Presentation	**Application**	Data format (part of the application)
5 – Session	**Application**	Dialogue control (part of the application)
4 – Transport	**Transport**	TCP
3 – Network	**Internet**	IPv4
2 – Data Link	**Network Access (Link)**	Ethernet II
1 – Physical	**Network Access (Link)**	Cable / signals

In TCP/IP, the top three OSI layers are combined into one Application layer, and the bottom two OSI layers are combined into one Network Access layer.

8. Alternative Concept Analysis – Industrial EtherType 0x8892

In this concept, the 12-byte command is placed directly inside an Ethernet frame using the industrial EtherType **0x8892**. The IPv4 and TCP headers are removed.

8.1 Frame size comparison

Item	Normal frame (Ethernet + IPv4 + TCP)	Direct frame (EtherType 0x8892)
Ethernet header	14 bytes	14 bytes
IPv4 header	20 bytes	– (removed)
TCP header	20 bytes	– (removed)
Command data	12 bytes	12 bytes
FCS	4 bytes	4 bytes
Total frame	**70 bytes**	**30 bytes**
Overhead percentage	58 ÷ 70 = 82.86 %	18 ÷ 30 = 60.00 %

8.2 Bytes saved per command

Bytes saved = 70 − 30 = 40 bytes
(equal to the IPv4 header + TCP header = 20 + 20)
Saving % = (40 ÷ 70) × 100 = 57.14 %

Answer: 40 bytes saved per command (57.14 % smaller frame)

8.3 Additional commands per second on a 100 Mbps link

First convert the link speed into bytes per second:

100 Mbps = 100,000,000 bits/s ÷ 8 = 12,500,000 bytes/s

Commands per second = link speed (bytes/s) ÷ frame size (bytes):

Normal frame: 12,500,000 ÷ 70 = 178,571.43 commands/s
Direct frame: 12,500,000 ÷ 30 = 416,666.67 commands/s
Additional commands/s = 416,666.67 − 178,571.43 = 238,095.24

Answer: about 238,095 additional commands per second (≈ 2.33 times as many)

Frame type	Frame size	Frame size in bits	Commands per second
Normal (Ethernet + IPv4 + TCP)	70 bytes	560 bits	178,571
Direct (EtherType 0x8892)	30 bytes	240 bits	416,666
Difference	40 bytes	320 bits	238,095

Table 10: Theoretical command rate (whole numbers).

Check using bits: 100,000,000 ÷ 560 = 178,571.43 and 100,000,000 ÷ 240 = 416,666.67

8.4 Important notes on this calculation

- The command-rate calculation is **theoretical**. It **excludes physical-layer encoding and inter-frame timing** (such as the preamble and the gap between frames), so a real link carries fewer frames per second.
- **Additional note:** Standard Ethernet needs a minimum frame of 64 bytes including the FCS. A 30-byte frame would have to be padded with 34 bytes in practice. If padding is counted, the direct frame becomes 64 bytes, giving 12,500,000 ÷ 64 ≈ 195,313 commands/s, about 16,741 more than the normal frame. The assignment's values give the 30-byte figure above, so the main answer uses 30 bytes.

8.5 Advantages and disadvantages of removing IP and TCP

Advantages	Disadvantages
• Saves 40 bytes in every command (frame reduced from 70 to 30 bytes). • Lower overhead, so more commands per second on the same link. • Less processing at each device (no IP or TCP handling), so lower delay. • Simpler, faster frames suit small, frequent control messages.	• **No routing:** without the IP header the frame cannot cross routers. It stays in one local network (one broadcast domain). • **No reliability:** TCP's acknowledgement, retransmission and ordering are lost, so lost frames must be handled by the application or the device. • **No ports:** several applications cannot be separated using port numbers. • Not compatible with normal IP networks and standard tools; only devices that understand EtherType 0x8892 can use it. • TTL protection against looping packets is lost.

Table 11: Removing IP and TCP.

Conclusion: Removing IP and TCP gives speed and efficiency inside a small, controlled local network, but the network loses routing and reliable delivery. The choice depends on whether the robot and controller are on the same local network and whether lost commands can be tolerated.

9. Summary of Final Answers

Question	Final answer
Routers crossed (64 − 61)	**3 routers**
Total bytes on the wire – robot command	**70 bytes**
Total bytes on the wire – camera packet	**1,458 bytes**
Overhead – robot command (58 ÷ 70)	**82.86 %**
Overhead – camera packet (58 ÷ 1458)	**3.98 %**
Direct EtherType 0x8892 frame size	**30 bytes**
Bytes saved per command	**40 bytes**
Commands per second – normal / direct (100 Mbps)	**178,571 / 416,666**
Additional commands per second	**≈ 238,095**

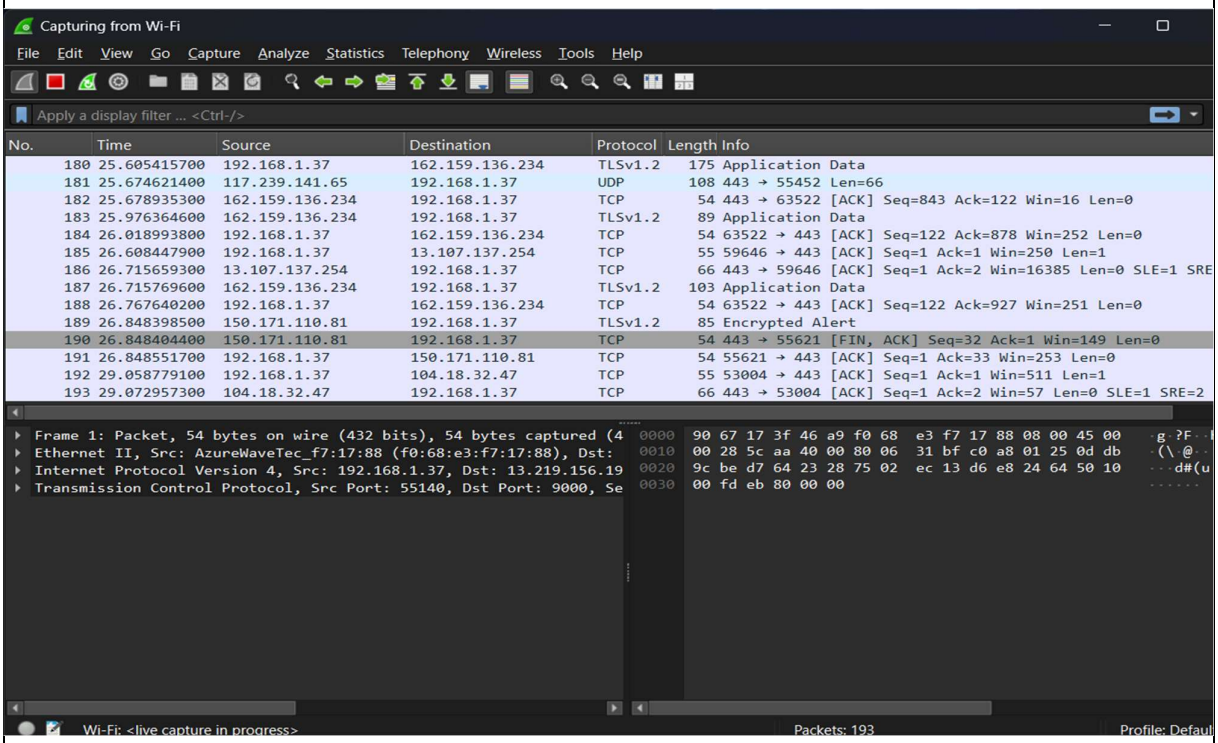
Table 12: Final answers.

10. Wireshark Practical Evidence to Be Attached

The assignment requires Wireshark evidence. This report does **not** contain any packet-capture screenshot, and it does not claim that a packet was captured on the student's computer. The values in Sections 2 to 8 are the values supplied in the assignment. The student must add his or her own captured evidence in the placeholders below.

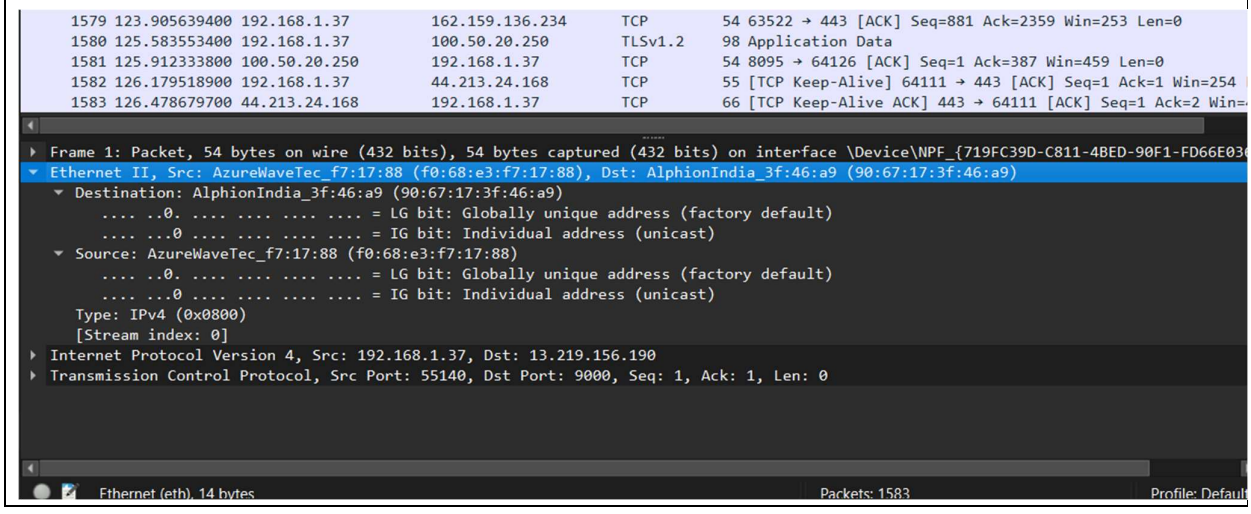
Evidence 1: Wireshark packet-list screenshot

Show the packet list with the selected frame highlighted.



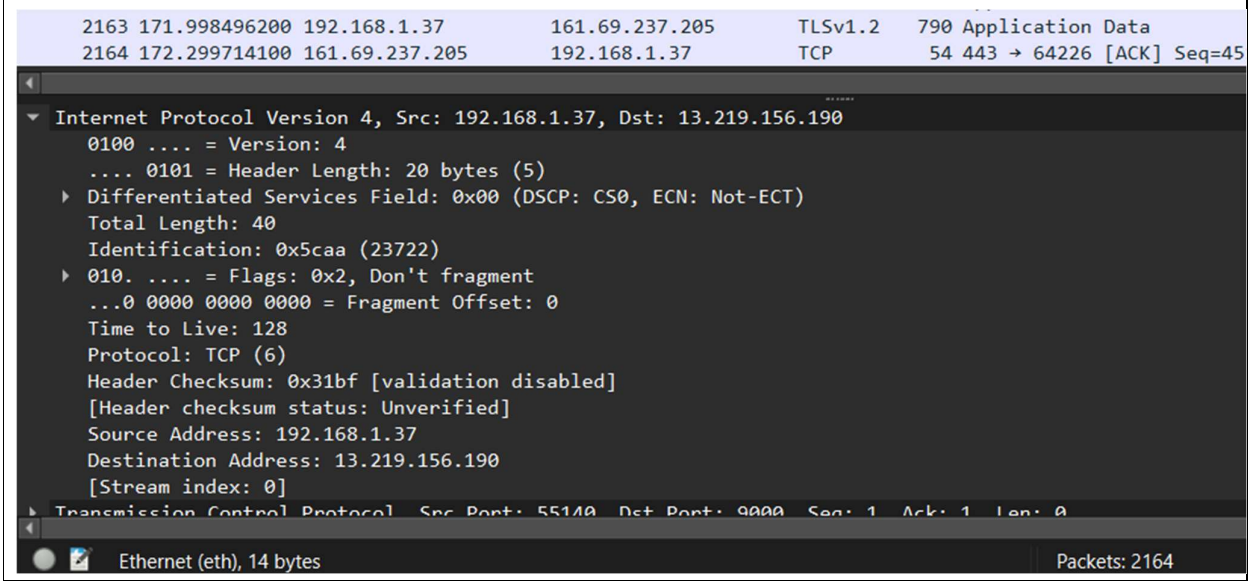
Evidence 2: Expanded Ethernet II header screenshot

Destination MAC, source MAC and EtherType visible.



Evidence 3: Expanded IPv4 header screenshot

Source IP, destination IP, TTL and protocol visible.



Evidence 4: Expanded TCP header screenshot

Source port, destination port, sequence number and acknowledgement number visible.

